

Measuring Physics-Driven and Data-Driven Contributions in Hybrid Process Models

Development and evaluation of a unique-contribution diagnostic metric for SO-PLS hybrid models of a two-stage batch reactor

Item	Value
Block structure	X1 -> M2 -> X2 -> Y (two-stage batch reactor)
Candidate mechanisms	M0 (correct) and six misspecifications M1-M6
Method	SO-PLS, sequential orthogonalisation, 10-fold CV (seed 42)
Calibration design	n = 100 batches, T2 ~ U(330, 370) K
Extrapolation design	n = 100 batches, T2 ~ U(370, 382) K
Report generated	17 September 2026

Headline result. The proposed metric — the ratio of the unique predictive contribution of the mechanistic block to that of the measured block — orders the seven candidate mechanisms perfectly against their out-of-domain performance (Spearman = -1.000), but with the *opposite sign* to the design hypothesis. Higher mechanistic dominance predicts *worse* extrapolation, not better.

1. Executive summary

This report documents the development of a calibration-only diagnostic metric intended to distinguish well-specified from misspecified mechanistic sub-models inside an SO-PLS hybrid, and its evaluation against genuine extrapolation data. Findings are reported as obtained, including those that contradict the design hypothesis.

1.1 What was established

- **The existing M2M metric has no intrinsic reference point.** Defined as $SS(M2)/SS(X2)$ in a single forward ordering, it credits all shared M2/X2 variance to M2. Observed values span 2.4 to 12.6 with no value carrying an interpretable meaning.
- **Shared variance dominates and cannot be attributed.** Commonality analysis gives unique contributions capturing only 17–37 % of the joint effect of M2 and X2; the shared term is 0.47–0.68 of a joint effect of ~ 0.84 in Q^2 units.
- **The proposed unique-ratio metric gains a natural reference at 1** and, on calibration data, separates mechanisms that omit a whole reaction pathway from those that do not.
- **The metric does not rank mechanisms by correctness.** M0 (correct) and M3 (wrong Ea4) are statistically indistinguishable on every estimator tried.
- **Calibration performance is blind to mechanism quality.** Across all seven candidates, within-domain R^2 spans only 0.8914–0.8978 — a range of 0.006.
- **The metric predicts extrapolation perfectly but inverted.** On regenerated data with corrected noise parameters and per-model fitted kinetics, $\text{Spearman}(\text{metric}, \text{extrapolation } R^2) = -1.000$ and $\text{Spearman}(\text{metric}, \text{degradation}) = -1.000$.
- **Stopping the flat-region rule at one standard error is justified;** going to two is not. The accuracy-for-parsimony exchange rate worsens threefold at the second step.

1.2 The central negative result

The metric measures *how heavily the hybrid leans on the mechanistic block*, and in this system that quantity is an inverse predictor of out-of-domain robustness. The asymmetry is structural: X2 consists of concentrations measured during stage 2 *under the test conditions*, so it is never out-of-domain for itself, whereas M2 is a fixed extrapolation of Arrhenius parameters estimated at $T_2 \in [330, 370]$ K and then evaluated at $T_2 \in [370, 382]$ K, where $\exp(-E_a/RT)$ amplifies any parameter error. A model that leans on M2 inherits that error in proportion.

This is corroborated independently at two levels. Across models the rank correlation is perfect. Within each of the seven models, across the 1-SE flat region, every additional M2 latent variable worsens extrapolation ($\rho = +0.28$ to $+0.69$) and every additional X2 latent variable improves it ($\rho = -0.56$ to -0.69) — fourteen consistent signs.

Note that mechanistic information still helps overall: every hybrid beat the pure data-driven benchmark on extrapolation (R^2 0.735–0.810 versus 0.633). The optimum is interior — some mechanistic content helps, heavy reliance hurts.

2. Data and regimes

Three distinct data regimes appear in this work. They are not interchangeable and absolute metric values are not comparable between them.

Regime	Source	Kinetic params	Noise sigma	Extrapolation set	Used for
A — real data	Author X1/X2/Y.xlsx	fitted per model	author (real)	none	metric development
B — DG generated	gen_calibration_extrapolation.py	fixed nominal	opt_sigma (incorrect)	yes (flawed control)	first extrap. test
C — regenerated	Data_Generation_corrected.py	fitted per model	author corrected	yes (proper control)	final extrap. test

Regime C was generated specifically for this report. The corrected diffusion coefficients are $\sigma_{\text{■}} = [0.75, 0.75, 0.25]$ and $\sigma_{\text{■}} = [0.25, 0.75, 0.25, 0.75, 0.75, 0.25]$, taken verbatim from the author's notebook; the previously used opt_sigma arrays came from a 1 %-CV calibration routine the author's notebook never runs.

A control-set defect was found and fixed. The Regime B "within-domain" test set spanned only T2 362–370 K, the top 20 % of the calibration range, which made extrapolation appear to *outperform* in-domain prediction. In Regime C the control is drawn from the same U(330, 370) distribution as calibration (99 % inside the calibration range) and the extrapolation set is 100 % outside it.

Set	n	T2 range (K)	Y mean +/- sd (%)	Role
RG_calibration	100	330.4 - 369.9	76.97 +/- 4.24	training
RG_within	100	330.5 - 370.0	76.36 +/- 5.08	in-domain control
RG_extrap	100	370.2 - 381.8	81.29 +/- 3.47	extrapolation

Fitted kinetic parameters, Regime C (least-squares against calibration X2 trajectories only):

model	Ea3	Ea4	A3	A4	fit_cost
M0	47828.5	53659.1	4.7	12.6	196,305
M1	50000.0	55000.0	11.4	20.0	3,001,417
M2	28251.8	41279.5	2.4	68.9	1,388,236
M3	48296.2	55000.0	5.5	2.0	200,229
M4	50018.0	55000.0	9.9	20.0	1,001,615
M5	12957.0	7379.1	9651.4	9670.0	36,227,650
M6	32428.4	55000.0	12.0	20.0	3,899,077

The fit recovers the true mechanism: M0 has both the lowest residual cost and near-true parameters (Ea3 47 829 versus 50 000; Ea4 53 659 versus 55 000). M5 is worse by two orders of magnitude.

3. SO-PLS block orderings and sequential sums of squares

Both orderings were run with an independent blind argmin over the full 6 x 15 x 15 grid. Sums of squares are sequential (Type I) on standardised Y, so SS_Y = 99 exactly and the decomposition closes to 0.000 for all fourteen fits.

3.1 Forward: X1 → M2 → X2

Model	LV X1	LV M2	LV X2	RMSECV	Q2	SS X1	SS M2	SS X2	SS F
M0	4	6	2	1.2641	0.9014	14.9229	71.7227	5.7065	6.648
M1	1	13	4	1.3156	0.8989	8.1626	73.7212	11.2673	5.8488
M2	4	6	4	1.2173	0.9102	14.9229	71.9646	6.194	5.9185
M3	4	6	2	1.2771	0.8987	14.9229	71.665	5.7293	6.6829
M4	4	8	2	1.2765	0.9028	14.9229	72.0508	5.7313	6.295
M5	3	7	4	1.3347	0.8912	14.3345	54.4125	22.9645	7.2885
M6	3	3	6	1.3666	0.8951	13.6644	65.9786	12.3664	6.9905

3.2 Reverse: X1 → X2 → M2 (independently re-optimised)

Model	LV X1	LV X2	LV M2	RMSECV	Q2	SS X1	SS X2	SS M2	SS F
M0	4	4	8	1.2722	0.9005	14.9229	61.5817	16.4267	6.0688
M1	2	5	13	1.2817	0.9021	12.265	65.7175	14.5112	6.5063
M2	1	6	4	1.2144	0.909	8.1626	72.2696	12.4326	6.1352
M3	4	3	8	1.2857	0.8985	14.9229	57.9766	19.7745	6.326
M4	4	4	7	1.2771	0.9004	14.9229	61.5817	16.2106	6.2848
M5	3	5	9	1.3802	0.8837	14.3345	66.8927	10.1269	7.6459
M6	1	9	1	1.357	0.8991	9.0382	72.3926	11.1889	6.3803

M0 reproduces the paper's {4, 6, 2} allocation and M6 its {3, 3, 6}. Predictive performance is essentially ordering-invariant (RMSECV differs by ≤ 0.04 and the sign is inconsistent), but the sequential SS is strongly order-dependent: for M0, SS(M2) falls from 71.72 to 16.43 when M2 is placed last, while SS(X2) rises from 5.71 to 61.58. Roughly 55–66 SS units (56–67 % of SS_Y) are jointly explainable and cannot be attributed by a single ordering. **This is the defect the new metric is intended to address.**

3.3 Two-block reference models

X1 → X2 contains no mechanistic block and is therefore model-independent.

Dataset	LV X1	LV X2	RMSECV	Q2	SS X1	SS X2	SS F
main dataset (M0-M5)	1	8	1.9211	0.7879	8.1626	73.7482	17.0892
mis dataset (M6)	6	6	2.0665	0.766	15.672	66.9032	16.4248

Model	LV X1	LV M2	RMSECV	Q2	SS X1	SS M2	SS F
M0	3	10	1.6008	0.8392	14.3345	73.1276	11.5379
M1	1	10	2.0283	0.7443	8.1626	71.4186	19.4188
M2	1	8	1.5884	0.8425	8.1626	78.7132	12.1241
M3	3	10	1.607	0.8375	14.3345	73.1037	11.5618
M4	1	13	1.6204	0.835	8.1626	79.1104	11.7269
M5	1	6	2.4647	0.624	8.1626	60.508	30.3294
M6	3	15	1.8962	0.8003	13.6644	71.9046	13.431

The mechanistic-only model out-predicts the data-driven one for every candidate except M1 and M5. M5 is the clear failure in isolation (RMSECV 2.4647, Q² 0.6240) — far more discriminating than the three-block result, where X2 compensates for the missing D-dependence.

4. Is the reported performance the best attainable?

Three separable questions were tested: whether the LV grid caps bound the optimum, how flat the RMSECV surface is around the argmin, and how much of the reported value is cross-validation partition luck.

Run	LV @cap15	RMSECV	LV @cap30	RMSECV	Gain	#within 1%	fold SE	seed mean	seed sd
A: X1->X2 main	(1,8)	1.9211	(1,8)	1.9211	0	1	0.1281	1.9474	0.0524
A: X1->X2 mis	(6,6)	2.0665	(6,6)	2.0665	0	5	0.1315	2.0801	0.0487
B: X1->M2 M0_correct	(3,10)	1.6008	(3,10)	1.6008	0	12	0.1962	1.6433	0.0495
B: X1->M2 M1_no_side	(1,10)	2.0283	(1,10)	2.0283	0	1	0.2387	2.1714	0.1035
B: X1->M2 M2_order1_D	(1,8)	1.5884	(1,8)	1.5884	0	7	0.1903	1.6114	0.031
B: X1->M2 M3_wrong_Ea4	(3,10)	1.607	(3,10)	1.607	0	12	0.1992	1.6525	0.0554
B: X1->M2 M4_lumped_EF	(1,13)	1.6204	(1,13)	1.6204	0	4	0.1998	1.6676	0.0528
B: X1->M2 M5_no_D	(1,6)	2.4647	(1,6)	2.4647	0	5	0.2847	2.5602	0.0431
B: X1->M2 M6_author_mis	(3,15)	1.8962	(3,15)	1.8962	0	32	0.142	1.9937	0.0648

No allocation was cap-limited. Doubling the latent-variable caps to 30 changed nothing anywhere, including the one case (M6) that sat on the boundary. The minima are genuine interior optima.

Seed 42 is a favourable partition. Across ten different cross-validation partitions every reported value sits at or below the ten-seed mean, by 0.01–0.14. Selection optimism from taking the argmin of a noisy surface is separately small (+0.003 to +0.066), so most of the offset is partition luck rather than over-searching.

Consequence for ranking. The M0 / M2 / M3 / M4 spread (RMSECV 1.588–1.620, range 0.032) is smaller than the seed-to-seed standard deviation of any one of them. Those four are not separated by this evidence; M1, M6 and especially M5 are separated by several standard deviations.

A structural ceiling exists. Y is computed from each batch's own final noisy sample, while X2 is truncated to a time grid shared across batches. The columns Y is an exact function of are therefore *not in* X2. This, not measurement noise alone, is what limits achievable accuracy.

5. Commonality decomposition

The proposal that the mechanistic contribution equals full – (X1+X2) and the measured contribution equals full – (X1+M2) is structurally correct, but these are *unique* (incremental) contributions and do not sum to the joint effect. Closing the decomposition requires the X1-only baseline.

Joint = $Q^2_{full} - Q^2_{X1}$ • Unique_M = $Q^2_{full} - Q^2_{DD}$ • Unique_X = $Q^2_{full} - Q^2_{KD}$ •
 Shared = Joint – Unique_M – Unique_X

Model	Q2 X1	Q2 KD	Q2 DD	Q2 full	Unique M	Unique X	sum	Joint	Shared
M0	0.0499	0.8392	0.7879	0.9014	0.1135	0.0621	0.1756	0.8514	0.6758
M1	0.0499	0.7443	0.7879	0.8989	0.111	0.1546	0.2657	0.849	0.5833
M2	0.0499	0.8425	0.7879	0.9102	0.1223	0.0677	0.19	0.8603	0.6703
M3	0.0499	0.8375	0.7879	0.8987	0.1109	0.0612	0.1721	0.8488	0.6767
M4	0.0499	0.835	0.7879	0.9028	0.1149	0.0678	0.1827	0.8529	0.6702
M5	0.0499	0.624	0.7879	0.8912	0.1033	0.2673	0.3706	0.8413	0.4707
M6	0.0586	0.8003	0.766	0.8951	0.1291	0.0948	0.2239	0.8365	0.6126

The two unique parts capture only 17–37 % of the joint effect. Shared is 0.47–0.68 out of a joint of ~0.84. The unique contributions are therefore floor estimates of each block's importance, not a partition of it.

Two structural observations shaped the metric design. First, Q^2_{DD} is identical (0.7879) for M0–M5 because the data-driven model contains no M2 block at all, so Unique_M is merely Q^2_{full} shifted by a constant and adds nothing as a comparative measure. Second, **Unique_X is the quantity that discriminates**, and it runs opposite to intuition: M5 (0.2673) and M1 (0.1546) stand far above the ~0.06 of the well-specified models. A large unique-X2 means X2 is doing work the mechanism failed to do.

The numbers depend heavily on protocol. Holding sub-models at the full model's latent-variable allocation instead of re-optimising roughly doubles Unique_M, because the data-driven baseline is starved of components (Q^2_{DD} collapses to 0.6562 for M0/M3/M4). The re-optimised protocol is the defensible one.

6. The proposed metric

$M2M_unique = unique(M2) / unique(X2)$, with a natural reference at 1: above 1 the mechanistic block contributes more unique predictive information than the measured block, below 1 the reverse. Four estimators were compared because they are *not* interchangeable.

Model	current M2M	SS-own	SS-match	SS-equal	CV mean	CV sd	P(>1)
M0	12.569	2.879	3.768	2.252	2.114	0.362	1.0
M1	6.543	1.288	1.487	1.186	0.697	0.106	0.0
M2	11.619	2.007	2.675	2.654	2.107	0.309	1.0
M3	12.509	3.451	3.751	2.226	2.09	0.397	1.0
M4	12.571	2.828	3.8	2.372	1.957	0.315	1.0
M5	2.369	0.441	0.642	0.617	0.375	0.035	0.0
M6	5.335	0.905	0.918	1.409	1.162	0.154	0.95

The reference at 1 works, under the CV estimator only. The two mechanisms that omit an entire chemical pathway, M1 and M5, fall below 1; the four that get the pathway structure right rise above it. This is unanimous across all twenty cross-validation partitions (P(>1) exactly 0.00 and 1.00). The current M2M cannot do this: its range is 2.37 to 12.57 with no anchor anywhere.

The in-sample SS estimators fail on M1 (1.19–1.49, wrongly above 1) because in-sample SS cannot decrease when latent variables are added, so the mechanistic side is inflated by however many components it was granted. The latent-variable asymmetry is large: M0's unique-M2 is measured with 8 components against 2 for unique-X2, and matching them moves the ratio from 2.879 to 3.769.

6.1 Choice of estimator

Most intuitive: the SS version — a variance split on a single deterministic fit, continuous with the existing M2M. But it is also the one that misclassifies M1. **Most interpretable of the working versions:** RMSECV, which reads in physical units ("adding M2 cuts prediction error by 0.65 purity points; adding X2 cuts it by 0.32"). **Preferred:** the Q^2 version, because only Q^2 differences decompose additively, so the Shared term means something; because it is more stable across partitions; and because, as Section 8 shows, it is far less sensitive to flat-region width.

Limitations carried forward. The metric separates {M0, M2, M3, M4} from {M1, M5} decisively but does not rank within either group: the spread (2.114, 2.107, 2.090, 1.957) is far smaller than the per-model standard deviation (0.31–0.40). M6, a genuine misspecification, lands above 1. The ratio is also unbounded above and undefined if $unique_X$ reaches zero.

7. Sensitivity to latent-variable allocation

Both estimators were swept over all 1 350 allocations per model, with sub-models held consistent with each candidate allocation.

Model	opt LV	SS @opt	SS median	SS P(>1)	Q2 @opt	Q2 median	Q2 P(>1)	Q2 undef.
M0	(4,6,2)	3.768	1.241	0.601	3.554	3.126	0.771	430
M1	(1,13,4)	1.487	0.666	0.277	1.033	1.212	0.591	4
M2	(4,6,4)	2.675	1.33	0.647	2.994	3.241	0.843	398
M3	(4,6,2)	3.751	1.233	0.599	3.643	3.021	0.757	448
M4	(4,8,2)	3.8	1.251	0.601	3.466	2.769	0.763	418
M5	(3,7,4)	0.642	0.337	0.072	0.751	0.622	0.062	0
M6	(3,3,6)	0.918	0.778	0.308	1.11	1.684	0.853	20

The threshold at 1 does not survive the full sweep. M0, the correct mechanism, falls below 1 at 40 % of allocations under the SS estimator. **The ranking does survive:** median ratio orders the models consistently under both estimators, with the four structurally sound mechanisms above the two known-broken ones.

The optimum is not a representative allocation. For M0/M2/M3/M4 the argmin sits well into the upper tail (M0: 3.77 at the optimum versus a median of 1.24). Quoting the ratio at the RMSECV optimum systematically flatters the mechanism.

Distribution of the unique-contribution ratio over all 1350 LV allocations ($X1 \leq 6$, $M2 \leq 15$, $X2 \leq 15$)

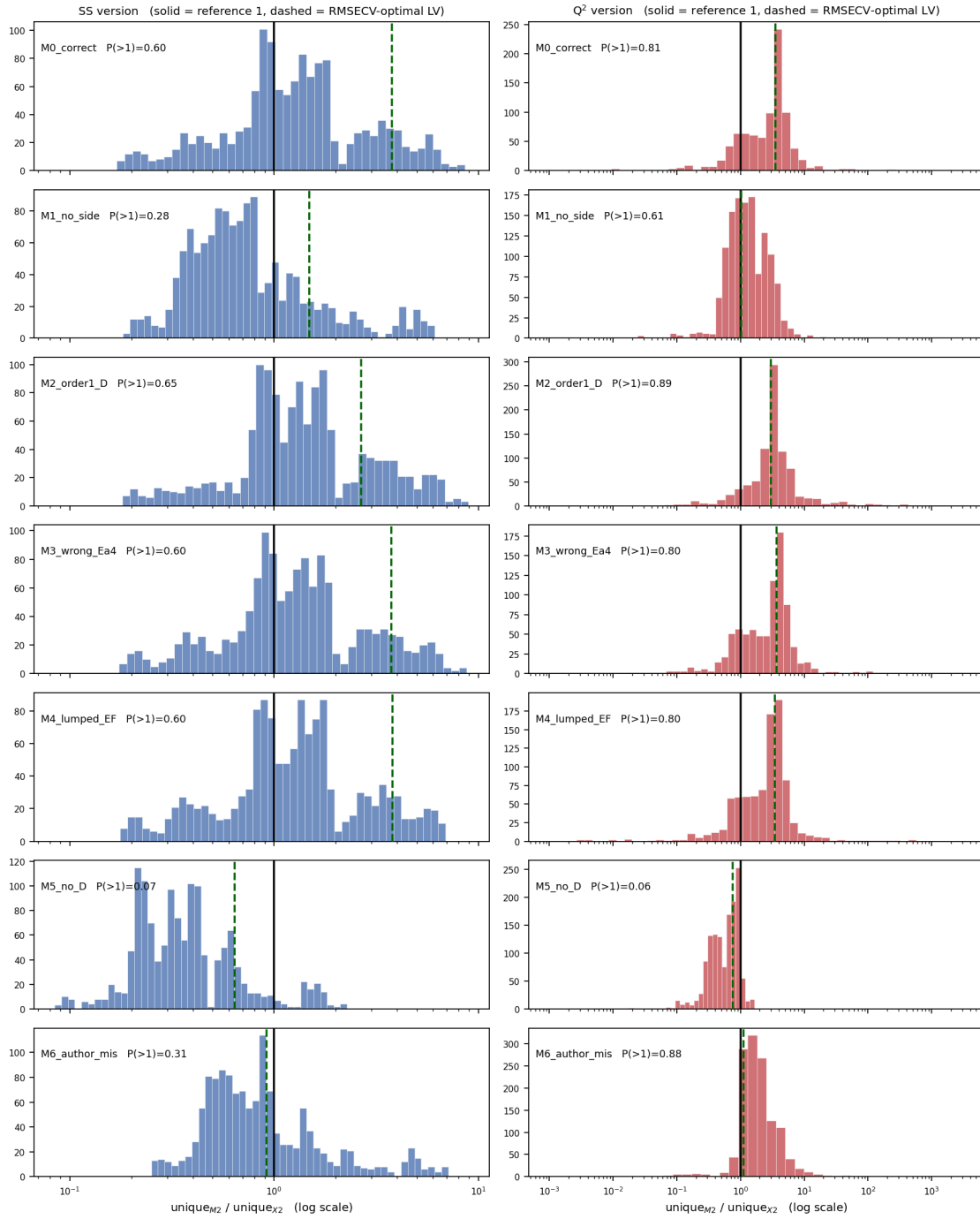


Figure 1. Distribution of the unique-contribution ratio over all 1 350 latent-variable allocations. Solid line: reference value 1. Dashed: RMSECV-optimal allocation.

8. Flat regions: 1-SE, 2-SE, 3-SE

8.1 Diagnostic power, $P(\text{ratio} > 1)$, Q^2 estimator

Model	1-SE	2-SE	3-SE	full grid
M0	0.983	0.827	0.845	0.771
M1	0.54	0.446	0.434	0.591
M2	0.994	0.913	0.919	0.843
M3	0.905	0.813	0.832	0.757
M4	0.904	0.826	0.828	0.763
M5	0.0	0.0	0.002	0.062
M6	0.742	0.839	0.874	0.853

8.2 Median ratio

Model	1-SE	2-SE	3-SE	full grid
M0	3.842	3.422	3.446	3.126
M1	1.014	0.948	0.924	1.212
M2	3.457	3.273	3.266	3.241
M3	3.842	3.556	3.59	3.021
M4	3.638	3.212	3.087	2.769
M5	0.66	0.471	0.448	0.622
M6	1.212	1.337	1.528	1.684

Restricting to the 1-SE region sharpens the classification dramatically (M0 goes from 0.60 to 0.98; M5 reaches exactly 0.000 under both estimators) and **eliminates the Q^2 estimator's numerical pathology entirely** — undefined cells fall from ~30 % of the full grid to 0.0 % at 1-SE and 2-SE. Q^2 is also far less sensitive to region width than SS: its medians move by 1.07–1.31x between 1-SE and the full grid, against 2.5–2.9x for SS.

Unique-contribution ratio within RMSECV flat regions (1-, 2-, 3-SE) vs the full grid



Figure 2. Ratio distributions within each flat region against the full grid.

8.3 The case for stopping at one standard error

The purpose of a flat-region rule is to trade accuracy for parsimony. Measuring that trade directly as percentage points of Q^2 surrendered per latent variable removed:

Model	LV saved (1-SE)	pp Q^2 / LV	extra LV (2-SE)	pp Q^2 / LV	x worse
M0	2	0.506	1	1.855	3.67
M1	7	0.249	2	0.651	2.61
M2	4	0.286	2	1.246	4.36
M3	2	0.459	2	1.576	3.43
M4	5	0.358	1	2.04	5.7
M5	3	0.65	2	0.74	1.14
M6	3	0.443	1	1.17	2.64

The first standard error buys simplification at **0.422 pp of Q^2 per latent variable**; the second buys it at **1.325 pp** — three times the price, for six of the seven models. It also removes 3.7 latent variables on average against 1.6 for the second step, so roughly 70 % of the achievable simplification happens in the first step.

Internal Q^2 spread of each region — is it still flat?

Model	1-SE	2-SE	3-SE	full grid
M0	0.024	0.05	0.08	0.3433
M1	0.0286	0.0444	0.0665	0.3314
M2	0.0232	0.0423	0.0682	0.3529
M3	0.0311	0.0585	0.0948	0.3411
M4	0.0233	0.0427	0.062	0.3433
M5	0.0266	0.0512	0.0764	0.294
M6	0.0212	0.0442	0.0692	0.343

At 1-SE the worst admitted model is 2.1–2.9 pp below the optimum; at 2-SE it is 4.2–5.6 pp. **A set whose members differ by nearly six points of Q^2 is not a flat region**, so widening to 2-SE does not merely cost accuracy — it invalidates the justification for using a region at all. Model ordering also degrades monotonically (Spearman against plain argmin: 0.750, 0.643, 0.464); by 3-SE, M1 — a mechanism missing a whole pathway — ranks first.

Honest counterweight. The 1-SE rule is not free: it costs 0.9–2.0 pp of Q^2 , or 1.8–3.2 standard deviations of partition noise. The 2-SE rule costs 4.4–6.8. The defensible claim is not that 1-SE is costless but that it is where the trade stops being favourable.

9. First extrapolation test (Regime B) and why it was repeated

The only pre-existing extrapolation data used fixed nominal kinetic parameters, incorrect noise coefficients, and the defective within-domain control described in Section 2. Every ratio in this regime fell below 1, so the reference threshold could not even be evaluated. Results are recorded for completeness.

Model	n 1-SE	RMSEP min	median	max	range	beat DD	rho LV_M2	rho LV_X2
M0	326	2.189	3.084	3.491	1.302	0.304	0.766	-0.521
M1	325	2.581	3.212	4.269	1.688	0.117	-0.038	0.342
M2	288	2.453	3.234	7.225	4.772	0.062	-0.086	0.295
M3	379	2.145	2.911	3.608	1.463	0.623	0.623	-0.306
M4	348	2.134	3.012	3.843	1.709	0.437	0.764	-0.45
M5	337	2.63	3.079	3.41	0.78	0.326	0.156	-0.086
M6	353	2.726	3.259	4.247	1.521	0.119	0.136	0.04

The within-model spread exceeded the between-model spread. Allocations that are statistically indistinguishable on calibration differed by 0.78 to 4.77 RMSEP units out of domain. Neither the argmin nor the 1-SE pick reliably found a good extrapolator: M3's argmin pick sat in the worst decile of its own region, while M6's 1-SE pick sat near its worst.

Crucially, the latent-variable direction result already appeared here: for the structurally sound mechanisms, $\rho(\text{LV_M2}, \text{RMSEP})$ was strongly positive (+0.62 to +0.77) and $\rho(\text{LV_X2}, \text{RMSEP})$ negative. **This regime and Regime C agree, which is what makes the finding credible.**

Extrapolation performance of every allocation inside the 1-SE flat region

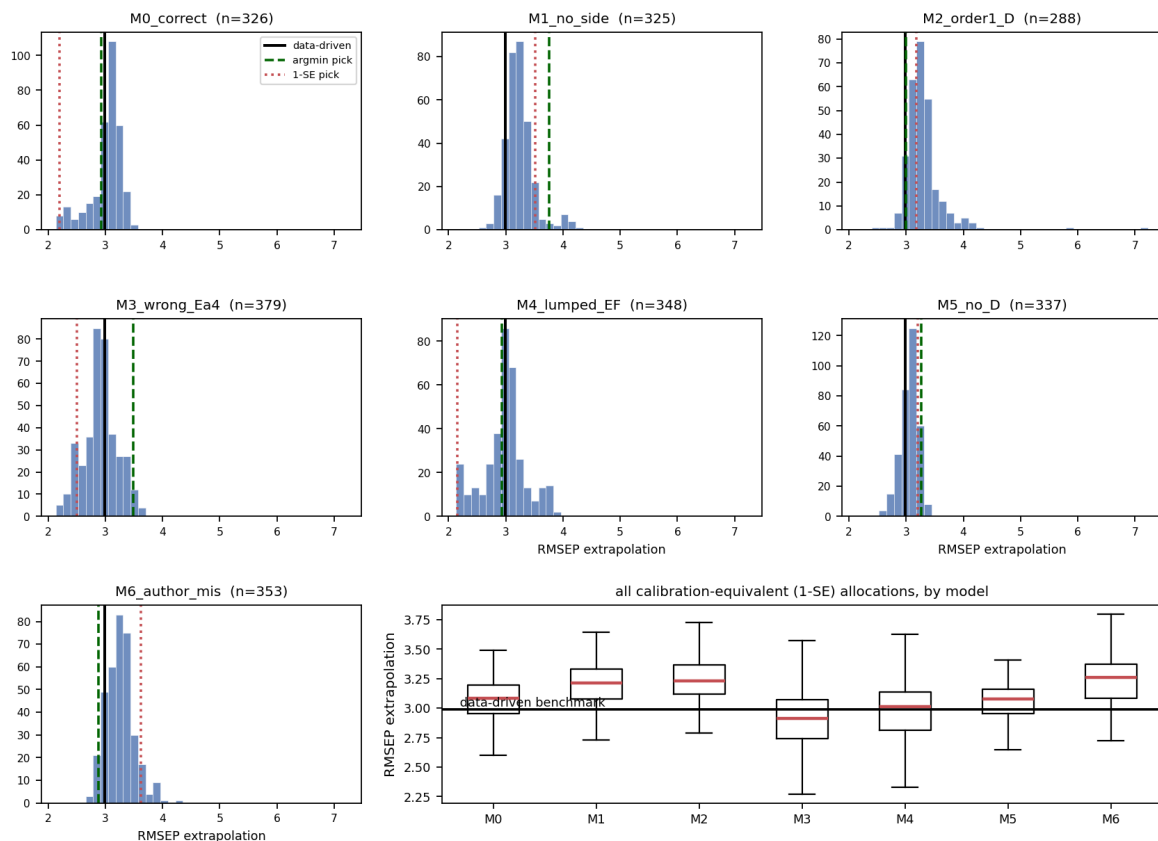


Figure 3. Regime B: extrapolation performance of every calibration-equivalent allocation.

10. Final extrapolation test (Regime C)

Regenerated data with the author's corrected noise coefficients, per-model kinetic parameters fitted on calibration only, and a proper in-domain control. For the first time all seven candidates sit on the same data, so M6 is numerically comparable to the rest.

10.1 Calibration side — the metric behaves as designed

Model	argmin LV	RMSECV	Q2 cal	n 1-SE	ratio Q2 med	p10	p90	P(>1)	ratio SS med
M0	(1, 11, 3)	1.1438	0.9198	232	4.6795	0.9124	7.7578	0.8621	3.2054
M1	(2, 3, 8)	1.131	0.9223	128	1.6614	0.8174	2.7721	0.8125	1.491
M2	(1, 9, 5)	1.212	0.91	314	3.0287	1.5848	6.8036	0.9459	2.1461
M3	(1, 11, 3)	1.1436	0.9197	257	4.7795	0.8559	7.6601	0.8366	3.381
M4	(1, 4, 6)	1.1632	0.9161	273	3.1279	0.8181	6.8299	0.8315	2.4856
M5	(4, 6, 5)	1.1579	0.9182	100	0.4958	0.3463	0.7308	0.0	0.4762
M6	(1, 7, 7)	1.1865	0.9141	231	1.533	1.0713	2.6482	0.9048	1.4037

The metric is again above 1 for six models with M5 the sole model below it, unanimously. Both estimators agree on the full ordering. Meanwhile calibration itself is blind: Q^2_{cal} spans 0.9100–0.9223 and within-domain R^2 spans 0.8914–0.8978.

10.2 Extrapolation — the inversion

Model	ratio Q2	R2 within	R2 extrap	dR2 (pp)	RMSEP extrap	beat DD
M3	4.7795	0.8916	0.735	-15.6618	1.7776	0.9844
M0	4.6795	0.8949	0.7464	-14.8494	1.7391	1.0
M4	3.1279	0.8953	0.7485	-14.6793	1.7317	0.989
M2	3.0287	0.8978	0.7819	-11.5886	1.6125	0.8885
M1	1.6614	0.8914	0.7851	-10.6311	1.6008	0.9609
M6	1.533	0.895	0.7922	-10.2861	1.5742	0.9481
M5	0.4958	0.8944	0.8101	-8.4239	1.5046	0.9

Spearman(metric, R^2_{extrap}) = -1.000. Spearman(metric, ΔR^2) = -1.000. Perfectly monotone across all seven models, for both estimators. The higher the mechanistic dominance, the worse the extrapolation and the larger the degradation.

The scale-free R^2 measure was used deliberately: the extrapolation set has lower Y variance (sd 3.47) than the within-domain set (sd 5.08), which flatters raw extrapolation RMSEP. The inversion survives normalisation.

It is not the metric specifically — it is mechanism reliance in general. Spearman(fit cost, R^2_{extrap}) = +0.964: the better a mechanism fits the calibration trajectories, the worse its hybrid extrapolates. M0, which recovers the true kinetics, degrades by 14.85 pp; M5, whose fitted parameters are nonsense, degrades by only 8.42 pp.

10.3 Within-model corroboration

Model	rho(ratio, RMSEP_e)	rho(LV_X1)	rho(LV_M2)	rho(LV_X2)
M0	0.66	-0.077	0.569	-0.693
M1	0.458	-0.064	0.281	-0.563
M2	0.566	-0.031	0.553	-0.616
M3	0.623	-0.048	0.501	-0.654
M4	0.672	-0.045	0.526	-0.675
M5	0.575	-0.089	0.686	-0.658
M6	0.43	-0.04	0.663	-0.636

Across the 1-SE region, for all seven models without exception, every additional M2 latent variable worsens extrapolation and every additional X2 latent variable improves it. Fourteen consistent signs from an independent level of analysis — considerably stronger evidence than the $n = 7$ rank correlation alone.

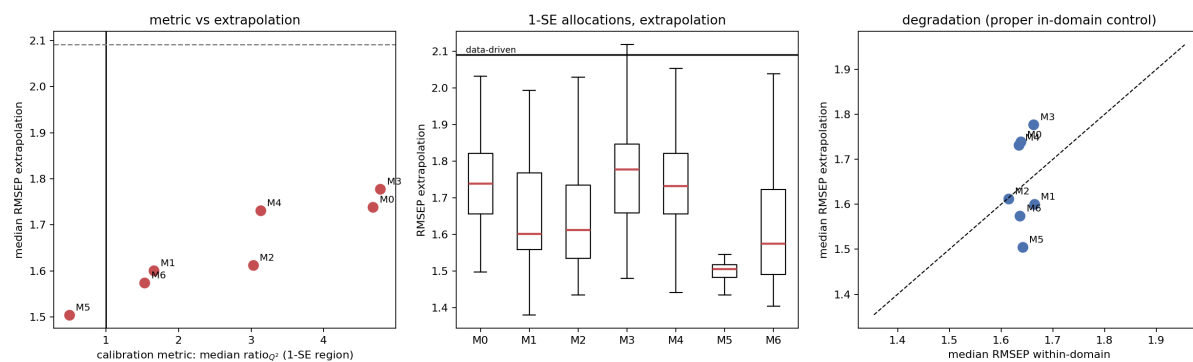


Figure 4. Regime C. Left: metric against extrapolation error (dashed grey = data-driven benchmark). Centre: extrapolation across all 1-SE allocations. Right: degradation against the proper in-domain control.

11. Conclusions

11.1 On the proposed metric

The unique-contribution ratio is a genuine improvement on the existing M2M in three respects: it does not credit shared variance to the mechanistic block, it has an intrinsic reference point, and it is well behaved numerically when evaluated over a 1-SE flat region. The recommended form is the cross-validated Q^2 version, reported as the median over the 1-SE region with a p10–p90 interval, alongside the Shared term so the reader can see how much the ratio does not account for.

It nevertheless fails at its intended purpose. It does not rank mechanisms by correctness: M3 (wrong activation energy) scores highest, M0 and M3 are indistinguishable on every estimator, and M5 — by far the worst mechanism — scores lowest while extrapolating best. What it measures is how heavily the hybrid leans on the mechanistic block.

11.2 On what the metric actually predicts

Read with the sign reversed, it is an excellent indicator of **extrapolation risk**. On Regime C it orders out-of-domain degradation perfectly. Whether this is a general property or an artefact of this block structure — where X2 is measured under the test conditions and therefore never out-of-domain for itself — cannot be settled by this data.

11.3 Limitations

- Regime C is a single realisation from a single seed. A rank correlation of -1.000 on seven points is striking but is one draw; the within-model evidence is what makes it credible. A second seed is the obvious confirmation.
- Parameter fitting used `max_nfev = 60`, matching the earlier canonical run. The hardest fits, M5 in particular, may not be fully converged.
- The metric values in Regimes A, B and C are not comparable; only within-regime comparisons are meaningful.
- The data-driven benchmark is a single allocation chosen by calibration Q^2 , compared against medians over hundreds of hybrid allocations. That specific comparison is indicative only.
- RMSECV (mean of fold RMSEs) and Q^2 (pooled PRESS) are not algebraically consistent; the allocation minimising one does not exactly maximise the other. The difference observed here is ≤ 0.03 pp.
- The finding that mechanistic reliance harms extrapolation may not transfer to designs where the measured block is also fixed at calibration conditions.

11.4 Recommended next steps

1. Repeat Regime C with a second and third generation seed to confirm the inversion. 2. Test whether the inversion persists when the mechanistic parameters are re-fitted using data that spans the extrapolation range, which would separate "mechanism is wrong" from "parameters were estimated in the wrong domain". 3. Investigate the fraction of allocations at which X2 adds nothing out-of-sample, which separated the sound from the broken mechanisms more cleanly than the ratio itself (0.295–0.332 versus 0.000–0.015) and was not pursued.

Appendix A. Analysis scripts and outputs

Script	Produces	Section
regen_corrected_data.py	RG_calibration / RG_within / RG_extrap .xlsx	2
regen_fit_and_test.py	regen_fit_and_test.xlsx + .png	10
sopls_two_orderings.py	sopls_two_orderings.xlsx	3
sopls_two_block.py	sopls_two_block.xlsx	3
sopls_two_block_ceiling.py	sopls_two_block_ceiling.xlsx	4
sopls_commonality.py	sopls_commonality.xlsx	5
metric_unique_m2m.py	metric_unique_m2m.xlsx	6
metric_ratio_lv_distribution.py	metric_ratio_lv_distribution.xlsx + .png	7
metric_ratio_flat_regions.py	metric_ratio_flat_regions.xlsx + .png	8
q2_flat_region_comparison.py	q2_flat_region_comparison.xlsx	8
extrapolation_test.py	extrapolation_test.xlsx + .png	9
extrapolation_1se_distribution.py	extrapolation_1se_distribution.xlsx + .png	9

Shared conventions across every analysis: mean-centring and unit-variance scaling fitted on calibration only, block scaling by $1/\sqrt{K}$, KFold(10, shuffle, random_state = 42), RMSECV as the mean of per-fold RMSE, $Q^2 = 1 - \text{pooled PRESS} / \text{SST}$, sequential sums of squares on standardised Y, and a latent-variable grid of $X1 \leq 6$, $M2 \leq 15$, $X2 \leq 15$.

Files not modified in the course of this work: Data_Generation.py, KD_fit.py, sopls_r.py. Data_Generation_corrected.py was created as a separate file and is imported unchanged; the only runtime interaction is setting its T2_range_testing attribute to carve the extrapolation range.